(a) Active exploration substantially boosts performance. Active agents exhibit emergent spatial strategies.

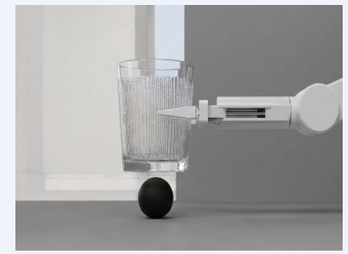
Question: is the chestnut in the glass?

GPT: yes. ✗
Gemini: yes. ✗

Initial

Strategy 1: move behind

Strategy 2: top down



Strategy 3: pick up

Strategy 4: pour down

(b) Action and perception failures cascade and compound. Bad actions result in bad views, which in turn bring worse actions

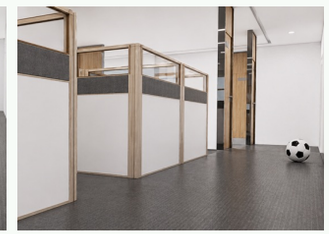
Question: which football is larger?

Initial

Move backward

Move up

Turn down



Move forward

Move forward for the rest steps.....

(c) Perception remains critically poor for specific tasks.

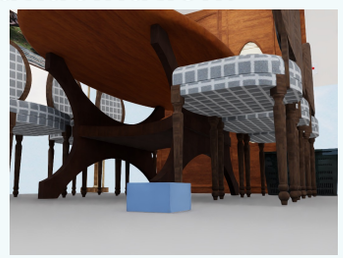
Question: Do the three form an equilateral, isosceles or random triangle?

GPT: random triangle ✗
Gemini: random triangle ✗

Question: which bottle corresponds to the one in the mirror?

A: I don't see a bottle in the mirror ✗

(d) Models exhibit premature commitment and spatial hallucination.



Question: Which box is closer to the center of the table? The blue one or black one?

A: Apparently the blue one.
Confidence = 1.0. Stop active exploring.

Question: which bottle corresponds to the one in the mirror?

A: : No. there's a clear gap.
Confidence = 0.95. Stop active exploring.